AQ10.1: Activity Questions 1 - Not Graded

This assignment will not be graded and is only for practice.
Level 1:

Consider a function $f: R^2 \rightarrow R$f : R2 → R defined as $f(x, y) = x^2 + 2xy^2 + 2x^2y + y^2$ f(x, y) = x2 + 2xy2 + 2x2y + y2. If $\nabla f(1, 1) = (a, b)$∇f(1, 1) = (a, b), then find the value of $a - b$a − b.

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 0
1 point
Consider the following function defined as

$$f_1(x, y)\text{f1(x, y)} = \begin{cases} \frac{x}{y} & \text{if } y \neq 0 \\ 1 & \text{if } y = 0. \end{cases}$$ {yx1if y = 0if y = 0.

$$f_2(x, y)\text{f2(x, y)} = xy + x^2y^2$$ xy + x2y2

- **Statement 1:** f_1 f1 has continuous gradient at the origin but f_2 f2 does not have continuous gradient at the origin.
- **Statement 2:** f_2 f2 has continuous gradient at the origin but f_1 f1 does not have continuous gradient at the origin.
- **Statement 3:** $f_1(x, y)$ and $f_2(x, y)$f1(x, y) and f2(x, y) have continuous gradients at the origin.

Which of the above statements is correct?

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 2
1 point
1 point
Consider a function $f: R^2 \rightarrow R$f : R2 → R defined as $f(x, y) = e^{(x^2 + xy + y^3)}$ f(x, y) = e(x2+xy+y3).

Which of the following option(s) is (are) true?

- [ ] $\nabla f = (e^{(x^2 + xy + y^3)}, e^{(x^2 + xy + y^3)})$∇f = (e(x2+xy+y3), e(x2+xy+y3))
- [ ] $\nabla f = (2x + y, 3y^2 + x)$∇f = (2x + y, 3y2 + x)
- [ ] $\nabla f = (2xe^{(x^2 + xy + y^3)}, 3y^2e^{(x^2 + xy + y^3)})$∇f = (2xe(x2+xy+y3), 3y2e(x2+xy+y3))
- [ ] $f(x, y)$f(x, y) increases most rapidly in the direction of $\frac{1}{5}(3, 4)$51(3, 4) at the point (1,1).
- [ ] $f(x, y)$f(x, y) decreases most rapidly in the direction of $\frac{1}{\sqrt{17}}(-1, -4)$ 17 √ 1(−1, −4) at the point (1,-1).

No, the answer is incorrect.
Score: 0
Accepted Answers:
$f(x, y)$f(x, y) increases most rapidly in the direction of $\frac{1}{5}$(3, 4)51(3, 4) at the point (1,1).
$f(x, y)$f(x, y) decreases most rapidly in the direction of $\frac{1}{\sqrt{17}}(-1, -4)$ 17

√

1(−1, −4) at the point (1,-1).

Level 2:

With a particular frame of reference (in R^2 R2), a corner of a hot rectangular iron plate with some finite length and breadth is placed at the origin (0, 0)(0, 0). The temperature of the plate at a point (x, y)(x, y) is given by the function:

$$T(x, y) = xy^2 + x^2y \quad \text{T(x, y) = xy2 + x2y}$$

Use this information to answer the following questions

1 point
From the point (1,3), in which direction the rate of change in temperature is the minimum?

○
$\frac{1}{\sqrt{274}}$ 274

√

1(-15, -7)

○
$\frac{1}{\sqrt{82}}$ 82

√

1(9, 1)

○
$\frac{1}{\sqrt{82}}$ 82

√

1(-9, -1)

○
$\frac{1}{\sqrt{274}}$ 274

√

1(15, 7)

No, the answer is incorrect.
Score: 0
Accepted Answers:
$\frac{1}{\sqrt{274}}$ 274

√

1(-15, -7)

1 point
From the point (1,3), in which direction the rate of change in temperature is the maximum?

○
$\frac{1}{\sqrt{274}}$ 274

√

1(-15, -7)

○
$\frac{1}{\sqrt{82}}$ 82
√
1(9, 1)
○
$\frac{1}{\sqrt{82}}$ 82
√
1(-9, -1)
○
$\frac{1}{\sqrt{274}}$ 274
√
1(15, 7)

No, the answer is incorrect.
Score: 0
Accepted Answers:
$\frac{1}{\sqrt{274}}$ 274
√
1(15, 7)

1 point

From a point (1,3), in which direction the rate of change in temperature of the iron plate is zero?

☐
$\frac{1}{\sqrt{274}}$ 274
√
1(-7, 15)
☐
$\frac{1}{\sqrt{82}}$ 82
√
1(1, -9)
☐
$\frac{1}{\sqrt{82}}$ 82
√
1(-1, 9)
☐
$\frac{1}{\sqrt{274}}$ 274
√
1(7, -15)

No, the answer is incorrect.
Score: 0
Accepted Answers:
$\frac{1}{\sqrt{274}}$ 274
√
1(-7, 15)
$\frac{1}{\sqrt{274}}$ 274
√
1(7, -15)

1 point

Consider a function $f: R^2 \rightarrow R$f : R2 → R defined as $f(x, y, z) = 2e^{2x} \sin yz$
f(x, y, z) = 2e2x sin yz. At the point (0, $\frac{\pi}{2}$, 12π, 1), which are the directions that have the directional

derivative to be 0 ?

$\frac{1}{\sqrt{2}}$ 2

$\sqrt{}$

1 (0,1,1)

$\frac{1}{\sqrt{2}}$ 2

$\sqrt{}$

1 (0,1,-1)

$\frac{1}{\sqrt{3}}$ 3

$\sqrt{}$

1 (1,1,1)

$\frac{1}{2021\sqrt{2}}$2021 2

$\sqrt{}$

1 (0,2021,-2021)

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\frac{1}{\sqrt{2}}$ 2

$\sqrt{}$

1 (0,1,1)

$\frac{1}{\sqrt{2}}$ 2

$\sqrt{}$

1 (0,1,-1)

$\frac{1}{2021\sqrt{2}}$2021 2

$\sqrt{}$

1 (0,2021,-2021)

1 point

Consider a function $f: R^3 \rightarrow R$f : R3 → R defined as $f(x, y, z) = \alpha x + \beta x y^2 + \gamma y z^2$ f(x, y, z) = αx + βxy2 + γyz2. If the maximum value of the directional derivative of $f(x, y, z)$f(x, y, z) at (1, 1, 1) is in the direction parallel to xx- axis and has magnitude 5, then

○ $f(x, y, z) = 5x + x^2$ f(x, y, z) = 5x + x2

○ $f(x, y, z) = 2x + x y^2$ f(x, y, z) = 2x + xy2

○ $f(x, y, z) = 5x$f(x, y, z) = 5x

○ $f(x, y, z) = x y^2 + 2 y z^2$ f(x, y, z) = xy2 + 2yz2

○ $f(x, y, z) = 2x$f(x, y, z) = 2x

No, the answer is incorrect.

Score: 0

Accepted Answers:

$f(x, y, z) = 5x$f(x, y, z) = 5x

Check Answers